

Environmental Product Declaration



In accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021 for:

TSM cement board

from

Zhejiang Terasun Air Duct Co.,LTD



Programme:

Programme operator:

EPD registration number:

Publication date:

Valid until:

The International EPD® System, www.environdec.com

EPD International AB

EPD-IES-0018268

2025-03-14

2029-03-14

An EPD should provide current information and may be updated if conditions change. The stated validity is therefore subject to the continued registration and publication at www.environdec.com



General information

Programme information

Programme:	The International EPD® System
Address:	EPD International AB Box 210 60 SE-100 31 Stockholm Sweden
Website:	www.environdec.com
E-mail:	info@environdec.com

Accountabilities for PCR, LCA and independent, third-party verification

Product Category Rules (PCR)

CEN standard EN 15804 serves as the Core Product Category Rules (PCR)

Product Category Rules (PCR): *PCR 2019:14 Construction products (version 1.3.4)*
UN CPC code: 37570

PCR review was conducted by:
The Technical Committee of the International EPD® System. The review panel could be contacted via info@environdec.com.

Life Cycle Assessment (LCA)

LCA accountability: *Sally Xie, Intertek*
sally.xie@intertek.com

Third-party verification

Independent third-party verification of the declaration and data, according to ISO 14025:2006, via:

☒ EPD verification by individual verifier

Third-party verifier: *Rui Wang, IVL Swedish Environmental Research Institute*
Approved by: The International EPD® System

Procedure for follow-up of data during EPD validity involves third party verifier:

☐ Yes ☒ No

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but registered in different EPD programmes, or not compliant with EN 15804, may not be comparable. For two EPDs to be comparable, they must be based on the same PCR (including the same version number) or be based on fully-aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have equivalent system boundaries and descriptions of data; apply equivalent data quality requirements, methods of data collection, and allocation methods; apply identical cut-off rules and impact assessment methods (including the same version of characterisation factors); have equivalent content declarations; and be valid at the time of comparison. For further information about comparability, see EN 15804 and ISO 14025.

Company information

Owner of the EPD:

Zhejiang Terasun Air Duct Co.,LTD

Contact:

info@terasun.cn

Description of the organisation:

Zhejiang Terasun Air Duct Co.,LTD., founded in 1989, is a large enterprise specializing in production of building products. Terasun is committed to researching and developing environmental-friendly and energy saving building materials.

Product-related or management system-related certifications:

ISO 9001:2015

Name and location of production site(s):

Zhejiang Terasun Air Duct Co.,LTD

(Maren) Chongren Industrial Clustering Area, Shengzhou City, Shaoxing, Zhejiang, China

Product information

Product name:

TSM cement board



Figure 1 Image of the product

Product description:

TSM cement board is made with cement, Talc powder, EPS foam particles and fiberglass mesh. It can withstand prolonged exposure to moisture in both of interior and exterior applications.

In this EPD, the nominal dimension of the studied TSM cement board is 12 mm thick, 1200 mm wide, and 2400 mm long.

UN CPC code:

37570

Geographical scope:

A1-A3 China; A4 From China to Italy; A5 Italy; C Italy; D Italy.

LCA information

Declared unit:

1 m² of TSM cement board product

The conversion factor is 9.96 kg per m² (0.100 m²/kg) based on the product of TSM cement board.

Time representativeness:

1st July 2023 to 30th June 2024 (12 months)

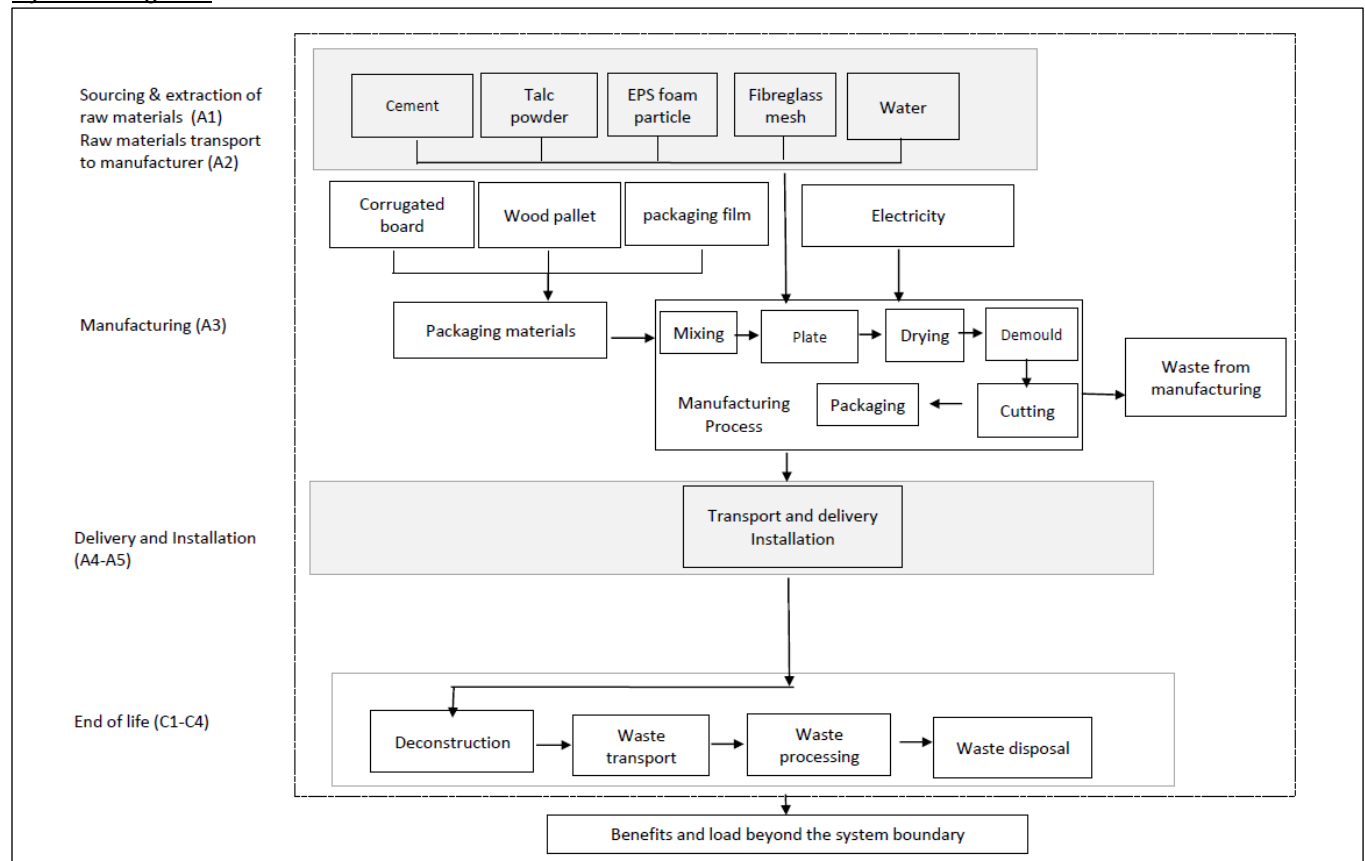
Database(s) and LCA software used:

Ecoinvent 3.10 (Allocation, cut-off by classification) and Simapro 9.6.0.1 software

Description of system boundaries:

The system boundary is type b) cradle to gate (A1-A3) with options (A4 and A5), module C1-C4, and module D. The life cycle stages are analysed in the study, including: A1-A3 product stage, A4-A5 construction process stage, C1-C4 end-of-life stage, and D benefits and loads beyond the system boundary.

System diagram:



Modules declared, geographical scope, share of specific data (in GWP-GHG results) and data variation (in GWP-GHG results):

	Product stage			Construction process stage		Use stage							End of life stage				Resource recovery stage
	Raw material supply	Transport	Manufacturing	Transport	Construction installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
Module	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Modules declared	X	X	X	X	X	ND	ND	ND	ND	ND	ND	ND	X	X	X	X	X
Geography	CN	CN	CN	CN to IT	IT	-	-	-	-	-	-	-	IT	IT	IT	IT	IT
Specific data used	13%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation – products	0%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation – sites	0%			-	-	-	-	-	-	-	-	-	-	-	-	-	-

More information:

Electricity data source used in the manufacturing process in A3: Market group for electricity, low voltage, CN-ECGC, in Ecoinvent 3.10 (cut-off)

Climate impact: 0.884 kg CO₂ eq./kWh

Product stage (A1-A3)

A1, Raw material supply takes into account the extraction and processing of all raw materials and energy which occur upstream to the studied manufacturing process. Specifically, raw material supply covers sourcing of cement, talc powder, EPS (expandable polystyrene) foam particles, and Fibreglass mesh.

A2, Transport to the manufacturer. The transportation of the raw materials to the manufacturing site is studied in this module.

A3, Manufacturing.

Mixing all the above raw materials with water to create a cementitious slurry. This slurry is then formed to wet panels and the fiberglass mesh is applied to both sides of the panel. These panels are then stored in the drying room under proper conditions to fully cure. The cured panels are then demoulded and are sent for cutting. The finished panels are stacked on the wood pallet with corrugated board on top and coner for protection, and wrapped around with packaging film. Electricity is consumed during the manufacturing process.

Quality checks are made at each step of the production process.

Packaging-related flows in the production process are included in the manufacturing module, i.e. packaging film, wood pallet and corrugated board. Apart from production of packaging material, the supply and transport of packaging material are also considered in the LCA model.

Construction process stage (A4-A5)

A4, Transport to the building site. This module includes transport from the production gate to the building site. Transport is calculated on the basis of a scenario with the parameters described. The average transportation distance from production plant to building site is 693 km transported by lorry and 8405 nautical miles (i.e., 15566.06 km) transported by ship.

A5, Installation into the building. The products covered by this EPD are designed to be mechanically fastened to steel framing or other framing members. The electricity usage for drilling screws and sawing panels is 0.025kwh per declared unit. Total 12 pieces of screws per declared unit are used, the total mass of the screws is 0.041kg, which is less than 1% of the mass input. The input of the screws is considered to be cut-off flow.

During installation, approximately 3-5% of the product is lost as off-cuts. For conservative calculation, 5% loss is taking for study. The additional production processes to compensate the loss is considered in this study. All product losses are collected for landfill disposal.

The impacts associated with packaging disposal are included with the installation phase. The packaging waste includes wood pallet, packaging film, and corrugated board box in A5. The end-of-life scenario of packaging is followed EU 27 waste management scenario.

Packaging	Recycling	Landfill	Incineration
Wood pallet	32.0%	38.4%	29.6%
Packaging film	40.7%	22.4%	36.9%
Corrugated board box	82.5%	9.3%	8.2%

End-of-Life Stage (C1-C4):

C1, De-construction. According to the owner, the product can be manually removed from the building/construction. Hence no impact is considered during demolition.

C2, Transport to waste processing. It is estimated that there is no mass loss during the use of the product, therefore the end-of-life product is assumed that it has the same weight with the declared product. All of the end-of-life product is assumed to be transported as separate construction waste to the closest facilities. Transportation distance to the closest disposal area is estimated as 100 km and the transportation method is lorry which is the most common.

C3, Waste processing for reuse, recovery and/or recycling. It is assumed 100% of the deconstructed products (C1) to be sent to landfill. Hence, no waste processing is required.

C4, Disposal. The 100% of the deconstructed products are assumed to be sent to landfill.

Resource Recovery Stage (D)

D, Reuse/recovery/recycling potential.

100% of the products are assumed to be sent to landfill.

No benefit or load resulting from reuse/recovery/recycling beyond the product system boundary.

CUT-OFF CRITERIA

The study does not exclude any modules or processes which are stated mandatory in the EN 15804:2012+A2:2019 and the applied PCR. The study does not exclude any hazardous materials or substances.

The study includes all major raw material and energy consumption. All inputs and outputs of the unit processes, for which data is available for, are included in the calculation. There is no neglected unit process more than 1% of total mass or energy flows. The module specific total neglected input and output flows also do not exceed 5% of energy usage or mass.

The input of the screws used in A5 was considered to be cut-off flow.

ALLOCATION

Allocation is required if some material, energy, and waste data cannot be measured separately for the product under investigation.

In this study, as per EN 15804, allocation is conducted in the following order;

1. Allocation should be avoided.
2. Allocation should be based on physical properties (e.g., mass, volume) when the difference in revenue is small.
3. Allocation should be based on economic values.

Allocation used in Ecoinvent 3.10 environmental data sources follows the methodology "allocation, cut-off by classification". This methodology is in line with the requirements of the EN 15804 standard.

For data sets in this study, the allocation of the inputs is generally carried out via the mass. The consumption and transportation of raw materials was allocated by mass ratio.

In this study one allocation occurs on product production, in allocating the input and output, i.e. energy within the production site such as electricity, and auxiliary material such as water, and waste water, among the various series of panel products, allocation is done via total production (panel area with the unit as m²) of all panel products produced on a yearly average.

During the production process of the product, the product scrap was used internally in the same product system for the products included in this study. This is a case of closed-loop recycling, thereby no co-products leave the product system, and no allocation was done for the product scrap, i.e. burdens of materials entered the final products are calculated in this study.

For this project, there is only one production site. So, there is no allocation among plants.

Key assumptions

1. 100 km transportation distance is assumed for the disposal of the product loss, packaging waste and the deconstructed products.
 2. The 100% of the deconstructed products are assumed to be sent to landfill.
 3. The dataset of the raw material talc powder was not available in Ecoinvent, silica sand, magnesium oxide and water from Ecoinvent database was used as proxy data for stoichiometric calculation.
- A sensitivity analysis was conducted.

Inclusion or exclusion of Infrastructure and/or Capital goods

Depending on the PCR, in general, the production and end-of-life processes of infrastructure or capital goods used in the product system should be excluded, unless there is evidence that they are relevant in terms of their environmental impact, or when a generic LCI dataset includes infrastructure/capital goods, and it is not possible, within reasonable effort, to subtract the data on infrastructure/capital goods from this dataset (directly citation from section 4.3.2 of PCR 1.3.4). In this study, the infrastructure and capital goods are not included in the LCA analysis since they are used plenty of times for several years for the product manufacturing. According to the PCR, it should be excluded.

Content information

Product components	Weight, kg	Post-consumer material, weight-%	Biogenic material, weight-% and kg C/declared unit
Cement	5.800	0%	0%
Talc powder	2.721	0%	0%
Fibreglass mesh	0.300	0%	0%
Water	0.996	0%	0%
EPS foam particles	0.143	0%	0%
TOTAL	9.960	0%	0%
Packaging materials	Weight, kg	Weight-% (versus the product)	Weight biogenic carbon, kg C/declared unit
Wood pallet	0.2500	2.5100%	0.0980
Corrugated board	0.0240	0.2410%	0.0102
Packaging film	0.0187	0.1878%	0
TOTAL	0.2927	2.9388%	0.1082

Substances, REACH – Very High Concern

The products do not contain any REACH SVHC substances in amounts greater than 0.1% (1000ppm).

Results of the environmental performance indicators

This LCA analysis applied the EN 15804+A2, EF3.1 was used

Mandatory impact category indicators according to EN 15804

Indicator	Unit	A1-A3	A4	A5	C1	C2	C3	C4	D
GWP-fossil	kg CO ₂ eq.	7.87E+00	2.40E+00	5.98E-01	0.00E+00	1.07E-01	0.00E+00	9.98E-02	0.00E+00
GWP-biogenic	kg CO ₂ eq.	-3.85E-01	3.80E-04	4.07E-01	0.00E+00	1.94E-05	0.00E+00	1.77E-04	0.00E+00
GWP-luluc	kg CO ₂ eq.	2.93E-03	1.13E-03	2.48E-04	0.00E+00	3.66E-05	0.00E+00	2.42E-05	0.00E+00
GWP-total	kg CO ₂ eq.	7.49E+00	2.40E+00	1.00E+00	0.00E+00	1.07E-01	0.00E+00	1.00E-01	0.00E+00
ODP	kg CFC 11 eq.	4.40E-08	3.77E-08	5.35E-09	0.00E+00	2.15E-09	0.00E+00	3.12E-09	0.00E+00
AP	mol H ⁺ eq.	3.12E-02	5.05E-02	4.35E-03	0.00E+00	3.45E-04	0.00E+00	1.10E-03	0.00E+00
EP-freshwater	kg P eq.	1.18E-03	1.06E-04	8.55E-05	0.00E+00	7.26E-06	0.00E+00	1.67E-04	0.00E+00
EP-marine	kg N eq.	6.82E-03	1.28E-02	1.09E-03	0.00E+00	1.17E-04	0.00E+00	2.74E-04	0.00E+00
EP-terrestrial	mol N eq.	7.29E-02	1.42E-01	1.15E-02	0.00E+00	1.27E-03	0.00E+00	2.95E-03	0.00E+00
POCP	kg NMVO C eq.	2.41E-02	3.99E-02	3.52E-03	0.00E+00	5.62E-04	0.00E+00	1.08E-03	0.00E+00
ADP-minerals&metals*	kg Sb eq.	1.09E-04	3.64E-06	5.95E-06	0.00E+00	2.88E-07	0.00E+00	1.91E-07	0.00E+00
ADP-fossil*	MJ	6.74E+01	3.11E+01	5.91E+00	0.00E+00	1.55E+00	0.00E+00	2.32E+00	0.00E+00
WDP*	m ³	1.28E+00	9.98E-02	-7.18E-03	0.00E+00	7.37E-03	0.00E+00	-1.30E+00	0.00E+00
Acronyms	GWP-fossil = Global Warming Potential fossil fuels; GWP-biogenic = Global Warming Potential biogenic; GWP-luluc = Global Warming Potential land use and land use change; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential, Accumulated Exceedance; EP-freshwater = Eutrophication potential, fraction of nutrients reaching freshwater end compartment; EP-marine = Eutrophication potential, fraction of nutrients reaching marine end compartment; EP-terrestrial = Eutrophication potential, Accumulated Exceedance; POCP = Formation potential of tropospheric ozone; ADP-minerals&metals = Abiotic depletion potential for non-fossil resources; ADP-fossil = Abiotic depletion for fossil resources potential; WDP = Water (user) deprivation potential, deprivation-weighted water consumption								

** Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator.*

Statement: The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks.

Additional mandatory impact category indicators

Indicator	Unit	A1-A3	A4	A5	C1	C2	C3	C4	D
GWP-GHG ¹	kg CO ₂ eq.	7.88E+00	2.40E+00	6.08E-01	0.00E+00	1.07E-01	0.00E+00	1.00E-01	0.00E+00

Resource use indicators

Indicator	Unit	A1-A3	A4	A5	C1	C2	C3	C4	D
PERE	MJ	6.33E+00	3.06E-01	4.24E+00	0.00E+00	2.39E-02	0.00E+00	4.66E-02	0.00E+00
PERM	MJ	3.66E+00	0.00E+00	-3.66E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	9.99E+00	3.06E-01	5.84E-01	0.00E+00	2.39E-02	0.00E+00	4.66E-02	0.00E+00
PENRE	MJ	6.11E+01	3.11E+01	6.70E+00	0.00E+00	1.55E+00	0.00E+00	7.85E+00	0.00E+00
PENRM	MJ	6.32E+00	0.00E+00	-7.94E-01	0.00E+00	0.00E+00	0.00E+00	-5.53E+00	0.00E+00
PENRT	MJ	6.74E+01	3.11E+01	5.91E+00	0.00E+00	1.55E+00	0.00E+00	2.32E+00	0.00E+00
SM	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	m ³	3.31E-02	3.12E-03	3.04E-04	0.00E+00	2.32E-04	0.00E+00	-2.80E-02	0.00E+00

Acronyms PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy re-sources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water

¹ This indicator accounts for all greenhouse gases except biogenic carbon dioxide uptake and emissions and biogenic carbon stored in the product. As such, the indicator is identical to GWP-total except that the CF for biogenic CO₂ is set to zero.

Waste indicators

Indicator	Unit	A1-A3	A4	A5	C1	C2	C3	C4	D
Hazardous waste disposed	kg	2.16E-03	5.87E-04	1.04E-03	0.00E+00	4.33E-05	0.00E+00	1.17E-03	0.00E+00
Non-hazardous waste disposed	kg	4.08E-01	9.74E-01	8.09E-01	0.00E+00	1.32E-01	0.00E+00	9.97E+00	0.00E+00
Radioactive waste disposed	kg	5.75E-05	5.30E-06	4.84E-06	0.00E+00	4.65E-07	0.00E+00	7.60E-07	0.00E+00

Output flow indicators

Indicator	Unit	A1-A3	A4	A5	C1	C2	C3	C4	D
Components for re-use	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Material for recycling	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Materials for energy recovery	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Exported energy, electricity	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Exported energy, thermal	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Disclaimer: it is discouraging the use of the results of modules A1-A3 (A1-A5 for services) without considering the results of module C.

References

General Programme Instructions of the International EPD® System. Version 4.0 and 5.0.

PCR 2019:14 Construction products, version 1.3.4

ISO 14025:2010 Environmental labels and declarations – Type III environmental declarations Principles and procedures.

ISO 14040:2006 Environmental management. Life cycle assessment. Principles and frameworks.

ISO 14044:2006 Environmental management. Life cycle assessment. Requirements and guidelines.

EN 15804:2012+A2:2019/AC:2021 Sustainability in construction works – Environmental product declarations – Core rules for the product category of construction products.

Ecoinvent 3.10 database.

Zhejiang Terasun Air Duct Co.,LTD background report. (version 2025-03-11)

